
1. Bank Account

Program Name: Bank.java

Input File: bank.dat

Judges Input File

```
2010-01-11 100.00
2010-01-30 100.00
2010-02-01 100324.32
2010-03-13 13243.53
2010-03-14 -1343.43
2010-06-11 -32.43
2010-11-11 1000.00
2010-12-31 100000.00
2011-01-30 1.00
2011-02-28 -13.33
2012-01-01 100.00
2013-01-02 1000.00
2014-02-23 -1000.00
```

Judges Output to Screen

\$505006.25

2. Blob Size

Program Name: BlobSize.java

Input File: blobsize.dat

Judges Input File

```
5
7 8 2
. . . . * * * .
. * . . . . *
. * * * . . . .
* * . * * . . .
. * * * . * *
. * * * . * *
. * * * . * *
* * . * * *
7 1
5 6
4 8 3
. . . . . * *
* * * . * * * .
. . . . . * *
* * * . * * * .
2 3
4 5
2 7
3 3 2
* * *
* . *
* * *
2 2
3 3
12 13 7
. * * . * * * * * *
* . . . * * . * * . *
. . * * . . . . * *
* . . . . * * . . . .
* * . * * * * . * *
* . * * * * . * *
* * . * * * . * *
* * . * * * . * *
* * * . * * * * *
* * * . * * * * *
* * . * * * . * *
* * . * * * . * *
* * . * * * . * *
* * . * * * . * *
* * . * * * . * *
4 3
7 6
12 2
3 3
8 13
12 13
3 11
4 12 3
* * * * . . . . * *
* * * . . . * . * *
. * * * . . . . * *
* . * * * * * * . *
2 7
1 5
4 1
```

2. Blob Size (cont.)

Judges Output to Screen

16
NO BLOB
3
10
10
NO BLOB
8
NO BLOB
16
19
18
19
3
NO BLOB
1
NO BLOB
27

3. DVD Elevator Algorithm

Program Name: DVDelevator.java

Input File: dvdelevator.dat

Judges Input File

```
10 2 200
100 300 400
900 210 220
1100 240 270
10000 220 240
10002 10000 20000
11003 30000 50000
20000 80000 90000
20002 20000 30000
200000 100 110
200100 130 131
210000 100 110
210300 120 122
300000 110 120
310000 500 700
310000 1000 1100
400000 10 30000
400001 9 11
```

Judges Output to Screen

```
Time 10011: 2 to 200
Time 10219: 210 to 220
Time 10229: 220 to 240
Time 10249: 240 to 270
Time 10309: 300 to 400
Time 20009: 10000 to 20000
Time 40009: 30000 to 50000
Time 90009: 80000 to 90000
Time 180009: 20000 to 30000
Time 239900: 100 to 110
Time 239920: 100 to 110
Time 239950: 130 to 131
Time 249962: 120 to 122
Time 310012: 110 to 120
Time 310402: 500 to 700
Time 310902: 1000 to 1100
Time 411091: 9 to 11
Time 411094: 10 to 30000
```

4. EXCELlent

Program Name: Excellent.java

Input File: excellent.dat

Judges Input File

```
20 17
SMITH, MARY
JONES, KERRY
ROBERTS, AMBER
JONES, JERRY
LASITER, RICK
ROGERS, GENE
ANDERSON, TROY
HARWOOD, SHIRLEY
GARCIA, RICARDO
GREEN, WES
LOCKE, RICK
VERDE, RICO
LETT, REX
MILLER, MATT
BENDER, BOB
RENC, LINDA
THOMAS, THOMAS
BRADFORD, BRAD
PEREZ, ANNA
CALDWELL, REX
SMITH, MARY
DOLE, ANN
JONES, KERRY
ROBERTS, AMBER
SMALLWOOD, LESTER
JONES, JERRY
ROGERS, GENE
RODRIGUEZ, JORGE
GARCIA, RICARDO
RENC, LINDA
THOMAS, THOMAS
RAWSON, BETH
BRADFORD, BRAD
LETT, REX
MILLER, MATT
NEALE, BOB
EAGLE, RIP
```

4. EXCELlent (cont.)

Judges Output to Screen

NAMES IN SPREADSHEET 1 THAT ARE NOT IN SPREADSHEET 2

ANDERSON, TROY

BENDER, BOB

CALDWELL, REX

GREEN, WES

HARWOOD, SHIRLEY

LASITER, RICK

LOCKE, RICK

PEREZ, ANNA

VERDE, RICO

NAMES IN SPREADSHEET 2 THAT ARE NOT IN SPREADSHEET 1

DOLE, ANN

EAGLE, RIP

NEALE, BOB

RAWSON, BETH

RODRIGUEZ, JORGE

SMALLWOOD, LESTER

5. Family Tree

Program Name: Family.java

Input File: family.dat

Judges Input File

Bob SON Mary
Josephus HUSBAND Mary
Brutus FATHER Mary
Vicky MOTHER Mary
Josephus SON Gary
Matilda MOTHER Josephus
Frank BROTHER Bob
Jessica WIFE Bob
Herman FATHER Jessica
Jessica DAUGHTER Gertrude
Nancy SISTER Jessica
Emanuel SON Nancy

Judges Output to Screen

1st Generation: Brutus Gary Matilda Vicky
2nd Generation: Gertrude Herman Josephus Mary
3rd Generation: Bob Frank Jessica Nancy
4th Generation: Emanuel

6. Frogger

Program Name: Frogger.java

Input File: frogger.dat

Judges Input File

```
7
8 8
*****..
....$.
..**2.
.3..**
**2.**
***.....
....@...
*****..
5 7
****$**
**....2
.*3..
.*...*
@..****
5 8
$...4...
..2...**
.3..2.**
**.....
*@...**
4 3
$1.
.**
...
..@
8 8
$*****
3***.
.....2
*****4
*.2.**2
.....**
.....
..12..@.
8 8
$..3..3*
.....2.
**...**3
***.....
2*2..2.
***...
***.....
***@****
4 5
$1*2.
...**
*.....
**.@.
```

6. Frogger (cont.)

Judges Output to Screen

8
6
6
5
9
6
4

7. Insertion Sort

Program Name: Insertion.java

Input File: insertion.dat

Judges Input File

```
7
8 3 5 7 2 A 4
5 3 7 2 9 T A 2 5
8 4 7 A K 2 T
2 A
2 3 9 T J Q K A
A K Q J T 8 6 4
3 6 8 3 9 A 3 Q J
```

Judges Output to Screen

```
3 8 5 7 2 A 4
3 5 8 7 2 A 4
3 5 7 8 2 A 4
2 3 5 7 8 A 4
2 3 5 7 8 A 4
2 3 4 5 7 8 A

3 5 7 2 9 T A 2 5
3 5 7 2 9 T A 2 5
2 3 5 7 9 T A 2 5
2 3 5 7 9 T A 2 5
2 3 5 7 9 T A 2 5
2 3 5 7 9 T A 2 5
2 2 3 5 7 9 T A 5
2 2 3 5 5 7 9 T A

4 8 7 A K 2 T
4 7 8 A K 2 T
4 7 8 A K 2 T
4 7 8 K A 2 T
2 4 7 8 K A T
2 4 7 8 T K A

2 A

2 3 9 T J Q K A
2 3 9 T J Q K A
2 3 9 T J Q K A
2 3 9 T J Q K A
2 3 9 T J Q K A
2 3 9 T J Q K A
2 3 9 T J Q K A

K A Q J T 8 6 4
Q K A J T 8 6 4
J Q K A T 8 6 4
T J Q K A 8 6 4
8 T J Q K A 6 4
6 8 T J Q K A 4
4 6 8 T J Q K A
```

(continued on next page)

7. Insertion Sort (cont.)

3 6 8 3 9 A 3 Q J
3 6 8 3 9 A 3 Q J
3 3 6 8 9 A 3 Q J
3 3 6 8 9 A 3 Q J
3 3 6 8 9 A 3 Q J
3 3 3 6 8 9 A Q J
3 3 3 6 8 9 Q A J
3 3 3 6 8 9 J Q A

8. MipMapping a Texture

Program Name: Mipmapping.java

Input File: mipmapping.dat

Judges Input File

```
4
4 4
0xffffffff 0xffffffff 0x00000000 0x00000000
0xffffffff 0xffffffff 0x00000000 0x00000000
0x00000000 0x00000000 0xffffffff 0xffffffff
0x00000000 0x00000000 0xffffffff 0xffffffff
8 8
0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc
0x7fcccccc 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc
7 8
0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc 0x7fcccccc
0x7fcccccc 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
0x7fcccccc 0xffffffff 0x00000000 0xffffffff 0xffffffff 0xffffffff 0xffffffff 0xffffffff
4 8
0x7faaaaaa 0xffff1111 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe
0x7faaaaaa 0xffffffff 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe
0x7faaaaaa 0xffffffff 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe
0x7faaaaaa 0xffff1111 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe 0xfefefefe
```

8. MipMapping a Texture (cont.)

Judges Output to Screen

```
0xffffffff 0x00000000
0x00000000 0xffffffff
0x7f7f7f7f

0x9fd89999 0xbfe56666 0xbfe56666 0x9fd89999
0xbfe5a5a5 0x7f3f3f3f 0x7f3f3f3f 0xbfe5a5a5
0xbfa5e5a5 0x7f003f7f 0x7f003f7f 0xbfa5e5a5
0x9f99d899 0xbf66e566 0xbf66e566 0x9f99d899
0xa7b87878 0xa7b87878
0xa769b888 0xa769b888
0xa7909880

NO REDUCTION POSSIBLE

0xbfd49999 0xfebebebe 0xfebebebe 0xbfd49999
0xbf99d499 0xfe7fbefe 0xfe7fbefe 0xbf99d499
0xdeaababb 0xdeaababb
```

Music CD

Program Name: Music.java

Input File: music.dat

Judges Input File

40
Can't You Hear Me Knocking 3 45
Sunshine of Your Love 4 12
While My Guitar Gently Weeps 4 23
Riders on the Storm 4 12
Back in Black 5 12
Smoke on the Water 3 56
Dance the Night Away 3 23
The Best Is Yet to Come 3 46
The Way You Look Tonight 3 32
Luck Be a Lady 4 21
Bewitched 3 42
The Good Life 3 56
The Girl from Ipanema 3 43
Fly Me to the Moon 2 34
Summer Wind 3 32
Strangers in the Night 3 52
Call Me Irresponsible 4 3
Somethin' Stupid 5 2
My Kind of Town 5 42
It Was a Very Good Year 5 21
That's Life 13 21
Maria 2 13
I Have Confidence 2 32
Sixteen Going On Seventeen 10 27
My Favorite Things 4 12
Do-Re-Mi 3 22
The Sound Of Music 2 12
The Lonely Goatherd 2 29
Edelweiss 2 34
Laendler 4 0
So Long, Farewell 8 0
Climb Ev'ry Mountain 4 0
My Favorite Things 4 0
Something Good 2 3
Processional And Maria 3 2
Sixteen Going On Seventeen 2 22
Do-Re-Mi 2 3
Edelweiss 3 12
So Long, Farewell 3 45
Climb Ev'ry Mountain 2 23

9. Music CD (cont.)

Judges Output to Screen

CD #1

Can't You Hear Me Knocking

Riders on the Storm

16:32

CD #2

Back in Black

The Way You Look Tonight

19:49

CD #3

Luck Be a Lady

Fly Me to the Moon

18:16

CD #4

Summer Wind

Somethin' Stupid

16:29

CD #5

My Kind of Town

It Was a Very Good Year

11:03

CD #6

That's Life

I Have Confidence

18:06

CD #7

Sixteen Going On Seventeen

Do-Re-Mi

18:01

CD #8

The Sound Of Music

So Long, Farewell

19:15

CD #9

Climb Ev'ry Mountain

Do-Re-Mi

17:30

CD #10

Edelweiss

Climb Ev'ry Mountain

9:20

10. Ones Galore

Program Name: Ones.java

Input File: ones.dat

Judges Input File

7 5 9 17 15 1 11 13 12 19

Judges Output to Screen

6

INVALID INPUT

9

16

INVALID INPUT

1

2

6

INVALID INPUT

18

11. Rings

Input File: rings.dat

Judges Input File

20
BBW
BBBW
BBWBB
BWBBB
BWWBBWB
B
W
WWBBBWBWW
WW
BBWBBBWBBW
WWBWWBWWWWBWWWB
WBBWWBBW
BWWBBWWB
WWWWBBB
WBBBBBWWWWBWBWBW
WWWWWWWWBBBBBWBWBWBBWWBWWWWBBBBBWW
BW
BBWB
WBBWB
BBWBWW

Judges Output to Screen

```

1 6
1 14
1 30
3 28
27 228
0 1
0 1
23 488
0 3
307 1740
1097 15286
51 204
51 204
7 120
1359 31408
1975983 1071765840
1 2
1 14
5 26
11 116

```

12. Roll

Program Name: Roll.java Input File: roll.dat

Judges Input File

7

```
EI LOPCOEV  TIMNUSCRCE*AE
TSOLVERIS E  HOBWPN0 LLMST UIITATE*
IPETER CER PAPPPCEDOI KIK L DECK PFRP PEPES*ABCDE
LU*I
EROCC EKTISISTN
IIF IT, MA  SSAISYI UTBID N*ATEFGHIC
NTO BE OR T TO BS TOEHAT SEITNQUHIT IHH TEEATKOD AVAY T  MAN BSDEIMES*A
EFGHIJKLC
```

Note #1: There is a space after the * in case #2.

Note #2: The last line in the input file appears as two lines in the printed version above.

Judges Output to Screen

```
I LOVE COMPUTER SCIENCE
SOLVE THIS PROBLEM TO WIN UIL STATE
PETER PIPER PICKED A PECK OF PICKLED PEPPERS
UIL
ROCKET SCIENTIST
IF IT IS MAY, IT IS AUSTIN
TO BE OR NOT TO BE THAT IS THE QUESTION THAT I HAVE ASKED MANY TIMES
```